

LACQUER: DECIDING WHAT TO FIX IN A FINISHED MIX

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ABSTRACT

Hand a finished mix to a mastering engineer and the first thing they do is nothing: they listen, they look at the meters, and only then do they reach for a tool. Most automatic restoration and mastering systems skip that step and map the input straight to new audio with one large model. Lacquer measures first and treats a damaged mix as separate questions. Clipped samples are rebuilt by consistent sparse reconstruction. A discrete echo is located in the cepstrum and removed by an exact inverse filter, unless its delay sits on the tempo grid, in which case it is music. Excess reverberation is removed by a band-split transformer that predicts a complex mask, and the same network, read as a meter, decides whether there was excess at all. Tone, image and dynamics are compared with the measured range of 103,838 released tracks, or with a reference, and moved no further than its edge. On held-out music the clipping and echo stages raise SI-SDR by 6.6 and 16.4 dB, the network gains 5.9 dB where four released models gain at most 0.7 dB, and 2.6 dB on rooms it never saw. We measure a bound on blind tone correction, beat Matchering on reference matching and on limiting at equal loudness and true peak, and leave 80 of 98 clean tracks bit-identical.

1. INTRODUCTION

A finished mix is one stereo file. The instruments, the effects, the room and whatever mastering it received are already folded together, and there are no stems to go back to. When something is wrong with it, a room that was too live, a delay printed by mistake, a converter that clipped, the only material to work with is the file itself. Recent systems answer this with a single generative model that maps damaged audio to clean audio [1, 2, 3], in the case of SonicMaster inside a pretrained audio latent. That formulation is general and appealing, and it has two costs. The whole output is synthesized, so where a latent decoder sits in the path its quality bounds the result even for the parts that needed no change. And one model has to treat a problem with an exact answer, a clipped sample, the way it treats a question of taste, such as stereo width.

Consider what the engineer does instead. They measure, and they sort. A discrete echo has an inverse; it is removed. Reverberation does not, but a model that has heard a great deal of music knows what the dry version should sound like; it is reduced. Level, tone and dynamics are not damaged in any absolute sense; they are compared with how released music of that kind usually measures, or with a reference the client brought, and left alone inside that

range. The decision comes before the tool, and a track that is already fine passes through untouched.

This paper describes a system built that way, and reports what we learned while building it. Some lessons are positive: a band-split masking transformer fine-tuned from a vocal dereverberation checkpoint is a strong full-mix dereverberator, and read as a meter it is a good judge of excess. Some are bounds: the spread of tone between released tracks is twice the size of a typical tonal fault, which caps blind equalization. And one is cautionary: our first version, run on clean released music, changed every track, because it mistook tempo-synced delays for echoes and quiet vocal samples for buried vocals. The final thresholds come from that audit. This work contributes:

1. **A pipeline that measures before it acts.** Nine stages for full-mix restoration and mastering at 44.1 kHz stereo, each reporting what it measured and what it changed (Section 3).
2. **A strong dereverberator and an honest number for unseen rooms.** The fine-tuned network beats WPE and four released models by a wide margin on its training reverb families; we report separately how much survives on rooms it never saw and how simulated rooms narrow the gap (Section 4.1).
3. **Meters, a tempo rule, and an audit.** Dereverberation models used as calibrated meters, a tempo-grid rule that separates echo faults from musical delays, and a do-no-harm audit of the whole repair chain on clean music (Sections 3.5 and 4.1).
4. **A measured bound on blind mastering, and what a reference buys.** Norms of 103,838 released tracks, a reference mode that beats Matchering on tone, and a true-peak limiter compared with open and commercial limiters at equal loudness (Section 4.4).
5. **Negative results with numbers, and one mechanistic observation.** Codec-token prediction, blind equalization, per-stem tone correction and de-compression fail in measured ways (Sections 4.4 and 4.5); the network’s time attention attends to the echo delay (Section 4.2).

Section 3 describes the stages in the order they run and Section 4 the experiments, including the negative ones. Code, the experiment log and audio examples accompany the paper.

2. RELATED WORK

Band-split models. Band-split RNN [4] and its transformer successors BS-RoFormer [5] and Mel-Band RoFormer [6] dominate music source separation. They split the complex spectrogram into bands, alternate sequence models over time and over bands, and predict complex masks. Community checkpoints trained with the

framework of Solovyev et al. [7] include dereverberation models for isolated vocals, which we use. Apollo [2] applies the idea to lossy-compressed music.

Generative restoration. SonicMaster [1] is the closest system to ours in scope: one flow-matching model in a pretrained audio latent that handles equalization, compression, reverberation and clipping in full mixes. Diffusion models have been applied to dereverberation [8] and to bandwidth extension and inpainting [3]. Richter et al. [9] show that models trained on room impulse responses underperform on artificial reverb, which is why artificial reverb is in our training mix.

Effects, mixing and mastering. A network can control an effect chain, through gradient approximation for black-box effects [10] or through differentiable effects [11, 12], and inference-time optimization adapts the processing to a reference [13, 14]. Blind estimation of effect parameters is hard because a good audio match does not require accurate parameters [15]; Moliner et al. [16] address blind equalization with a content-aware predictor. De-limiting has been cast as sample-wise gain inversion [17]. Removal of one or several effects from single-instrument recordings is studied by Rice et al. [18]. Released music has been measured before: its long-term spectrum [19], the loudness war’s effect on peaks and dynamics [20], and the level of the lead vocal in mixing studies [21] and across seven decades of popular music [22]. Reference mastering matches a target’s spectrum and level to a reference track [23], and ITO-Master [14] transfers a mastering style through a learned embedding. A commercial limiter applied to MUSDB18-HQ is available as musdb-XL [24].

3. METHOD

Let $x \in \mathbb{R}^{2 \times N}$ be a stereo mix at 44.1 kHz. Figure 1 shows the nine stages applied in order. Each returns audio and a record of its measurement and decision.

3.1. Training pairs

Clean music comes from FMA [25] and MUSDB18-HQ [26] (mixtures and stems). Damage is applied on the fly so that the damaged signal stays sample-aligned with its clean target. Reverberation adds a wet tail to the dry signal at a direct-to-reverberant ratio (DRR) of -6 to 12 dB; tails come from measured rooms [27], from band-wise exponentially decaying noise, and from an algorithmic comb-and-allpass design. Echo adds one to five taps at 60 to 500 ms. Clipping, noise at 20 to 45 dB SNR, parametric equalization, a feed-forward compressor, section-wise level changes and brickwall limiting complete the set. Segments carry two seconds of leading context so that tails of earlier audio fall inside the training window. Input and target are level-normalized, and 15% of pairs are undamaged, because a system that must leave clean audio alone has to see clean audio.

3.2. Clipping by consistent reconstruction

Hard clipping leaves exact knowledge behind. Samples below the ceiling θ are untouched, and samples at the ceiling were at least that large. A channel is treated as clipped when runs of samples sit within 10^{-3} of its peak on both polarities and make up more than $2 \cdot 10^{-4}$ of the signal. The clipped samples are then rebuilt with A-SPADE [28]: for each windowed frame (2048 samples, hop 512, DFT redundancy 2) it searches for the sparsest spectrum whose

waveform equals the input on the reliable samples and exceeds $\pm\theta$ on the clipped ones, relaxing the sparsity until the constraint residual is below a tolerance. A channel without a flat ceiling is left unchanged.

3.3. Echo by cepstral detection and exact inversion

A single echo is $y[n] = x[n] + a(g*x)[n-d]$ with delay d , gain a and a short echo-path filter g . Its spectrum is $Y(\omega) = X(\omega)(1 + aG(\omega)e^{-j\omega d})$, so the log-magnitude spectrum gains a ripple of period $1/d$ in frequency, and the real cepstrum gains a narrow peak at quefrency d whose shape is ag to first order [29]. We average the log spectrum over 3 s frames, take the cepstrum, and score each quefrency between 40 and 650 ms by its deviation from a running median in units of a running median absolute deviation. A peak with $z \geq 45$ is accepted; on clean held-out music the largest peak has median $z \approx 12$ and 95th percentile 39, while echoed clips have a 5th percentile of 62. The echo is then removed with the recursive filter

$$\hat{x}[n] = y[n] - \sum_k h_k \hat{x}[n-d-k], \quad (1)$$

where h is read from the cepstrum around the peak and refined by fixed-point iteration on the residual cepstrum. Because the recursion reaches back at least d samples it runs block by block. The procedure repeats until no peak remains, which handles multi-tap and feedback delays. One check comes first. If d is within 2% of a whole number of sixteenth notes or triplet eighths at the estimated tempo, the echo is almost certainly a delay effect or a loop, and it is left in place.

3.4. Masking network

The network is a BS-RoFormer [5] with 12 layers, width 256, 8 heads and 62 bands, 51 M parameters. The stereo STFT (window 2048, hop 512) $X \in \mathbb{C}^{2 \times 1025 \times T}$ is split into bands that are 2 bins wide at low frequencies and up to 129 bins at the top. Each band, with real and imaginary parts of both channels, is projected to a token, giving a grid of $T \times 62$ tokens; each layer applies attention along time within a band and then along bands within a frame, with rotary positions. A per-band head predicts a complex mask M and the output is

$$\hat{S} = M \odot X, \quad (2)$$

followed by an inverse STFT. The input passes around the network, so whatever the mask leaves at one comes back exactly, and there is no codec or vocoder in the path. Training minimizes

$$\begin{aligned} \mathcal{L} = & \lambda \|\hat{s} - s\|_1 + \sum_r \left(\|\hat{S}_r - S_r\|_1 \right. \\ & \left. + \mu \left\| \log(|\hat{S}_r| + \epsilon) - \log(|S_r| + \epsilon) \right\|_1 \right) \end{aligned} \quad (3)$$

over five STFT resolutions r (windows 4096 to 256), with $\lambda = 10$, $\mu = 0.5$. The log-magnitude term gives weight to reverberant tails, which are quiet. We initialize from a publicly released BS-RoFormer trained for vocal dereverberation and fine-tune on full mixes with AdamW, batch 4, 4 s segments, learning rate 10^{-4} with cosine decay, and an exponential moving average of the weights.

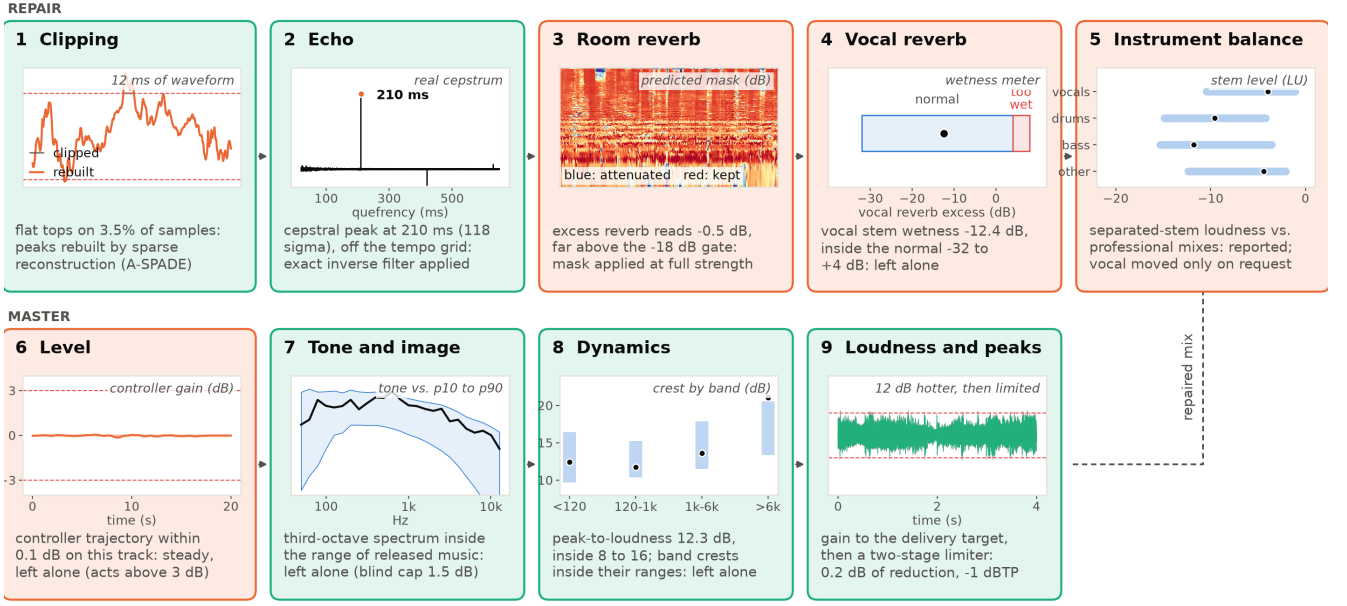


Figure 1: The decision pipeline, drawn from one 4 s clip with room reverb and a 210 ms echo added. Each card shows the quantity a stage measures and the decision it reached. Green cards are deterministic DSP; orange cards use the learned model.

3.5. Measuring before acting

A dereverberation model f gives a measurement for free. We define the wetness of a signal as the level of what the model removes relative to what it keeps, over active frames:

$$w_f(x) = 20 \log_{10} \frac{\|x - f(x)\|}{\|f(x)\|} \quad [\text{dB}]. \quad (4)$$

Excess reverberation of the mix. The full-mix network was trained to remove reverberation that was added to finished productions, so w measures reverberation *in excess* of a normal production. On MUSDB18-HQ training mixtures it reads a median of -41 dB as produced and -20 , -9 and -4 dB when reverberation is added at DRRs of 12, 6 and 0 dB, ordering these cases correctly in 97 to 100% of songs. The pipeline evaluates w on three windows, bypasses the network below a gate τ , and above it blends the restored signal with strength $\min(1, (w - \tau)/8)$. Released music of all genres reads higher on this meter (95th percentile -20.6 dB), so $\tau = -18$ dB is set from that population rather than from the training set.

Vocal ambience. Reverberation on a vocal is partly an artistic choice. We separate stems with HT-Demucs [30] and apply w with the vocal dereverberation model to the vocal stem, which measures absolute wetness. Produced vocals span a wide range (10th to 90th percentile -32 to -1 dB), so we act only on clear cases: above $+4$ dB, the 95th percentile on released music, the vocal is blended toward the dry estimate until it reads 0 dB; a dry vocal is reported, and a plate is added only on request; readings above $+8$ dB mean the stem is not a voice the meter understands. Stems are remixed only when the vocal was changed.

3.6. A controller that outputs parameters

Level, tone and dynamics are corrected by a second network that cannot generate audio. From mid and side log-mel features it pre-

dicts a static equalization curve $e(f)$ at 32 log-spaced frequencies and a gain trajectory $g(t)$ per frame, applied as the mask

$$20 \log_{10} G(t, f) = e(f) + g(t), \quad (5)$$

which is separable in decibels. The ideal parameters are available in closed form from a clean and damaged pair as band-wise and frame-wise level ratios, which serves both as an upper bound and as a training target. We train with a magnitude-only multi-resolution loss plus a Huber loss on the parameters, and a three-stage curriculum from one fault per clip to all faults at once. At inference the gain trajectory is computed on overlapping 8 s windows, offset so that windows agree where they overlap, clamped to ± 6 dB, held through quiet passages, and applied only when it exceeds 3 dB.

3.7. Mastering against measured norms

The mastering stages are deterministic and follow the same rule as the repair stages: measure, compare with a normal range, act only outside it. A track is described by integrated loudness and loudness range [31], true peak, peak-to-loudness ratio (PLR), a third-octave long-term spectrum, the side-to-mid level in five bands, the inter-channel correlation below 150 Hz, and the crest factor in four bands. The normal range of each feature is its 10th to 90th percentile over released music of the chosen genre, or over all genres.

Tone and stereo image. Third-octave bands and band-wise side levels outside the range are moved to its edge with one linear-phase filter. Without a reference a move is capped at 1.5 dB, the size mastering engineers describe as normal, and a reading further outside is reported as a mix problem rather than fixed. The low band is narrowed only when its correlation falls below 0.5, and no filter is applied as routine.

Dynamics. A band whose crest factor exceeds the range, or a mix whose PLR does, receives 2:1 compression on the excess

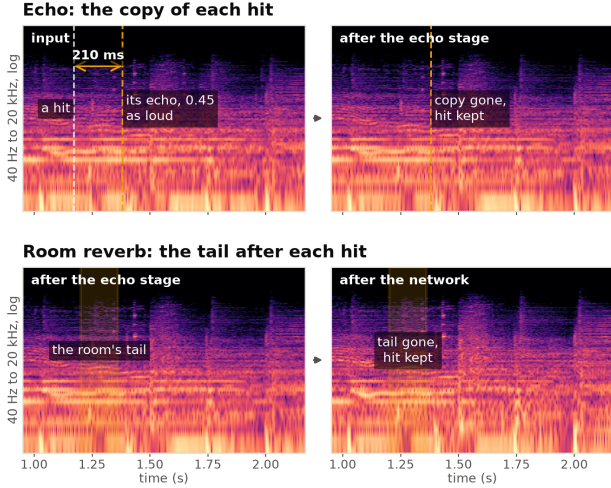


Figure 2: Before and after on 1.2 s of the clip of Figure 1: the echo stage removes the copy of each hit 210 ms later (top), the masking network removes the room’s tail after each hit (bottom).

(25 ms attack, 250 ms release); a mix that is already dense is not compressed.

Loudness and peaks. Gain to a delivery target [32] is followed by a limiter with a slow stage for sustained reduction, a fast lookahead stage, and a $4\times$ oversampled soft clipper that takes the last 1.5 dB off the shortest peaks. The ceiling is -1 dBTP, or -2 dBTP above -14 LUFS, the lower ceiling EBU Tech 3343 gives for data-reduced delivery [33]; the switch point is our own rule. The limiter has a budget of 3 dB of reduction; a target that needs more is not reached, and the shortfall is reported.

Reference. A reference track collapses every range onto the reference’s own features, which turns the same chain into reference mastering with larger permitted moves; tone and image are then matched by fitting the mid and side third-octave spectra separately.

Instrument balance. With stems separated, the loudness of each stem relative to the mix is compared with its range in professional mixes, measured on separated stems. A reading outside it is reported, and on request the vocal is moved to the edge of its range as a difference, $y = x + \sum_s (\tilde{s} - \hat{s})$, so that an untouched stem contributes nothing.

4. EXPERIMENTS

Data and metrics. Training uses FMA-medium [25] and the training split of MUSDB18-HQ [26]; evaluation uses held-out FMA tracks (split by a hash of the file name) and the MUSDB18-HQ test split. We report scale-invariant SDR (SI-SDR) [34] and log-spectral distance (LSD) against the clean clip, production quality from the reference-free Audiobox Aesthetics predictor [35], and signal descriptors; PESQ and STOI are speech metrics and are not used.

4.1. Restoration

Where the gain comes from. Table 1 is the main comparison, and its clearest message is about initialization. The released vocal models damage full mixes, as one would expect of models trained

on isolated vocals. Yet their weights are the best starting point we found: with the architecture held fixed, the same 51 M network trained from random weights reaches 4.0 dB on reverberant mixes after eight thousand steps, less than the fine-tuned one reaches after one thousand (6.4 dB), and no better than a 7.7 M model or a spectrogram U-Net. At this budget the warm start decides the result and the modules do not. At 23k steps the fine-tuned network gains 5.4 dB on reverberation. Clipped clips never gain SI-SDR from the network even though their spectra move closer to clean (LSD 10.6 to 4.9 dB), which is why Section 3.2 exists.

Rooms the network never saw. Table 2 compares the network with weighted prediction error (WPE) dereverberation [38], the standard blind method from speech, and with four released community models (Reverb_HQ, UVR-DeEcho-DeReverb and MDX23C-De-Reverb from the UVR model repository [39], and the vocal BS-RoFormer trained with [7]), run with the audio-separator package (version 0.47) at default settings. Set A is 20 held-out clips with reverberation from all three training families. Set B is 40 held-out clips with rooms no model here was trained on: 67 binaural impulse responses from six rooms of the Aachen Impulse Response database, version 1.4, which extends [40]. On set A the best released model gains 0.7 dB and ours 5.9 dB. On set B ours is the only method that improves the average, and it improves all 40 clips, but the gain shrinks to 1.7 dB (2.7 dB at DRR below 1 dB, 0.8 dB above 5 dB). The training rooms are short (median T_{30} 0.35 s) and monophonic; the Aachen rooms are longer (median 0.93 s) and binaural. The network had learned its 270 training rooms very well and rooms in general much less well. Adding 4000 simulated shoebox rooms (pyroomacoustics 0.10 [41], image sources plus its later ray tracing stage, a spaced microphone pair, T_{60} 0.25 to 2.8 s) as a second impulse-response source from step 16.5k raises the set B gain to 2.6 dB at step 23k (95% bootstrap interval 2.0 to 3.3 dB, better on 37 of 40 clips) with the validation score unchanged. On 250 such clips the Fréchet audio distance [42, 43] to 1500 clean training clips falls from 0.114 to 0.086 with CLAP [44] (clean test clips: 0.081) and from 0.74 to 0.23 with VGGish (clean: 0.21). All released models raise the log-spectral distance, mostly by thinning the mix.

Clipping. On 29 held-out clips hard-clipped at the 90th to 99.7th percentile of sample magnitude, the declipper of Section 3.2 raises SI-SDR from 20.1 to 26.7 dB and improves every clip (median $+6.2$ dB), while the network leaves it at 19.9 dB. The two do different jobs: the declipper restores the waveform, the network lowers the log-spectral distance (11.4 to 4.2 dB) by removing distortion products above the music’s bandwidth, and the two in sequence give 24.7 dB and 3.9 dB. The detector fires on 29 of 29 clipped clips, on none of the clean clips, and on none of 150 full-length MUSDB18-HQ mixtures. Tanh saturation has no flat ceiling, is not detected, and passes through unchanged.

Echo. On 79 held-out clips of 25 s with echoes of 60 to 500 ms, gain 0.15 to 0.6 and one to five taps, the cepstral stage of Section 3.3 raises SI-SDR from 8.8 to 25.2 dB (median 23.2): 9.4 to 29.2 dB for single taps and 7.9 to 19.0 dB for multi-tap echoes. It detects 71 of 79 and fires on 3 of 79 clips without added echo, at delays that suggested a tempo-synchronized delay or an exact loop in the music, the first hint of the problem the clean-music audit below made unmistakable. Under reverberation it detects 35 of 40 and gains 16 dB. The network alone moves echo from 9.5 to 12.3 dB.

Mix-level against stem-level decisions. On 24 MUSDB18-HQ test songs, room reverb on the whole mix (input 4.7 dB) is

Table 1: SI-SDR (dB) on a fixed validation set of 32 held-out tracks. “Identity” is the SI-SDR of the output against a clean input.

Model	steps	reverb	echo	reverb+echo	clip	identity
Damaged input		2.9	9.5	0.9	19.6	
Vocal BS-RoFormer, as released	0	−0.6	0.9	−4.8	3.6	5.3
Vocal Mel-Band RoFormer, as released	0	−16.7	−5.6	−18.5	−3.7	−2.4
Spectrogram U-Net, from scratch	8k	4.1	8.9	1.9	17.3	25.4
with CBAM [36]	8k	4.0	9.1	1.6	18.3	30.0
with CBAM and FiLM [37]	8k	4.0	9.1	1.7	18.2	28.9
Band-split transformer, from scratch (7.7 M)	8k	3.7	8.9	2.3	17.8	27.7
BS-RoFormer (51 M), from scratch	8k	4.0	9.1	2.4	17.7	25.8
BS-RoFormer (51 M), fine-tuned	1k	6.4	9.6	3.9	18.2	29.1
BS-RoFormer (51 M), fine-tuned	8k	7.6	11.7	5.7	19.1	35.9
BS-RoFormer (51 M), fine-tuned	23k	8.4	12.6	6.5	19.8	42.1

Table 2: Reverberant full mixes, clips of 12 s, DRR −3 to 9 dB. Set A: 20 clips, training reverb families. Set B: 40 clips, measured rooms unseen by every model. SI-SDR / LSD in dB.

Method	Set A	Set B (unseen rooms)
Damaged input	3.2 / 5.5	3.1 / 4.4
WPE [38]	3.0 / 5.2	0.9 / 4.6
Vocal BS-RoFormer	2.9 / 13.5	0.1 / 13.8
Reverb HQ (MDX-Net)	3.9 / 12.8	3.2 / 12.5
UVR DeEcho-DeReverb (VR)	3.4 / 11.2	3.1 / 9.9
MDX23C De-Reverb	−0.5 / 12.9	0.5 / 12.4
Ours, step 15k	9.1 / 4.2	4.8 / 4.1
Ours, with simulated rooms, 23k		5.7 / 4.0

best removed from the mix (9.8 dB, against 5.3 per separated stem, because the excess meter reads much lower on stems), while reverb on the vocal only (6.7 dB) is best handled on its stem (7.7 against 7.3 dB). Neither wins both, so the pipeline decides at two levels and reaches 9.7 and 7.8 dB, altering one of 24 clean songs.

Reference-free quality and musical descriptors. The Audiobox Aesthetics predictor [35], which never sees the clean clip, rates production quality on 24 held-out clips at 6.66 clean, 6.37 with reverberation and 6.63 restored (restored above damaged in 79% of clips); for clipping it sees no improvement, in agreement with the flat SI-SDR. On 16 held-out tracks the restored clips also move back toward the clean clip in musical terms: under reverberation the onset-envelope correlation rises from 0.949 to 0.971 and the estimated note decay time returns from 0.37 to 0.27 s (clean 0.28 s); under echo the correlation rises from 0.912 to 0.986.

Clean music. Any action on an undamaged track is an error, and this test taught us more than any other. We ran the repair stages, with stems, on 98 held-out released tracks of all genres. With the first thresholds no output was identical to its input: 13 tempo-synced delays were removed as echoes, the excess meter fired on 10 tracks, a quiet separated “vocal” (a sample or separation residue) was boosted on 21, and the controller moved every track slightly. With the tempo-grid check, thresholds set on released music, balance reported instead of applied, and level riding only above 3 dB, 80 of 98 outputs are bit-identical to their inputs; the rest are 5 echo removals, 3 reverb reductions, 4 vocal reverb reductions and 8 level rides. The tempo-grid check has a price, and we state it plainly: of 100 added echoes it removes 69 instead of 94, because a quarter of random delays fall on the grid, while it alters 3 clean clips instead of 11. With one stage removed at a time on 30 clips carrying an unseen room, an echo and clipping at

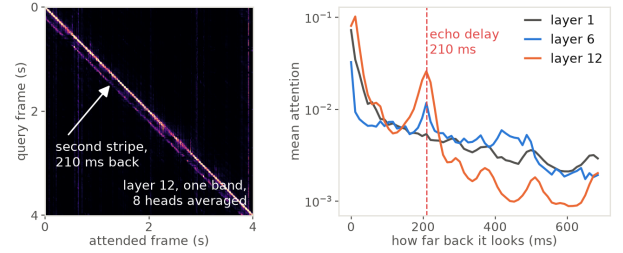


Figure 3: Time attention in one band for a clip with a 210 ms echo (18 frames). Left: layer 12 shows a second stripe 18 frames below the diagonal. Right: mean attention against lag; layers 6 and 12 peak at the echo delay.

once (input 2.3 dB), the full chain reaches 5.4 dB, 5.0 without the declipper, 4.8 without the echo stage and 3.5 without the network, so each stage carries its own fault.

4.2. What the network attends to

On a clip with a 210 ms echo (Figure 3), the time attention of the last layer shows a second stripe parallel to the diagonal at exactly that lag. To test whether this is general rather than a pretty picture, we added an echo with a random delay between 90 and 500 ms to 30 held-out clips and compared the lag profile of time attention (mean weight against frames back, over bands and heads) with the profile on the same clip without echo. In layer 1 the weight at the true lag is unchanged (ratio 1.00). In layer 6 the largest relative increase falls within one frame of the true delay in 30 of 30 clips (median ratio 2.5), and in layer 12 in 26 of 30 (median ratio 14.3). The delay is never given to the model. It has learned to find the earlier copy of each sound, which is the computation an echo canceller is built on.

4.3. Level controller

With ideal parameters the separable mask repairs nearly all of the damage in its family (level faults 3.67 to 0.01 dB). Learning the parameters blind is another matter. With an ℓ_1 loss on the parameters the network outputs exactly zero correction for thousands of steps, because the many frames that need no correction pull every prediction toward zero with constant force. A Huber loss removes that. With it, training on all faults from the start reaches an envelope error of 1.76 dB on level faults (input 3.67) while disturbing clean

Table 3: Error against a track’s own features after mastering a faulted copy (200 held-out tracks per fault, dB). Blind: population or genre norms. Reference: the original is given as reference.

Fault (measure)	damaged	blind		reference	
		all	genre	ours	Matchering
Tilt (tone)	3.13	3.03	2.98	0.19	0.72
EQ bumps (tone)	2.56	2.53	2.48	0.12	0.65
Width (side level)	5.68	5.09	4.93	0.27	0.03
Peakiness (band crest)	0.61	0.52	0.54	0.32	0.52
None (tone moved)	0	0.65	0.66	0.05	

clips by 0.43 dB, and the curriculum reaches 1.52 dB at 0.15 dB; a VU-ballistics rider reaches 2.53 dB at 0.11 dB. On 20 full 30 s clips the learned trajectory reduces the envelope error from 4.52 to 2.36 dB against 2.94 dB for the rider.

4.4. Mastering

What there is to correct. We measured the features of Section 3.7 on 103,838 released tracks [25] and on the 150 MUSDB18-HQ mixtures, which are professional mixes that were never mastered. The two populations have nearly the same spectral slope (-5.3 and -5.1 dB per octave, median) and side-to-mid level (-10.6 and -10.0 dB), and differ in dynamics: PLR 11.9 dB (10th to 90th percentile 8.0 to 16.3) against 15.8 dB (13.7 to 17.9) for the unmastered mixes. A good mix already has the right tone; what mastering adds is mostly dynamics and loudness.

The bound on blind correction. To 200 held-out released tracks we applied one mastering fault each (a spectral tilt of 0.6 to 1.6 dB per octave, one or two broad equalizer bumps of 3 to 8 dB, or a side-level change of 3 to 9 dB) and measured the error against the track’s own original features after mastering (Table 3). Population norms recover 1 to 3% of a tonal fault and genre labels add 2 points, and the reason is measurable. A held-out track’s own third-octave spectrum is 5.2 dB RMS from the population mean, 4.6 dB from its genre mean and 4.3 dB from the mean of its 20 nearest neighbours in descriptors that an equalizer does not change, while the fault is 2.3 dB: the fault is smaller than the spread it has to be told apart from. A Gaussian posterior-mean estimator with the full band covariance does no better, and a reference-free quality predictor [35] scores the original above the faulted version in only 47 to 60% of 800 comparisons. We therefore cap blind moves at 1.5 dB and treat anything stronger as a job for a reference.

Reference mastering. With the original track as reference, the chain of Section 3.7 removes 94 to 95% of a tonal fault. Against Matchering 2.0 [23] with the same reference it ends closer in tone on 80% of tilted and 88% of equalized tracks and closer in band dynamics; Matchering is closer on stereo width and on peak-to-loudness ratio (0.21 against 0.44 dB, because our output is held at 0 dBTP when the reference peaks above full scale). ITO-Master [14], its released white-box model run without its optional inference-time optimization, moves these clips away from the reference (tone error 3.2 to 7.8 dB on 19 tracks); it transfers a style through a fixed effect chain rather than matching a spectrum. On its own task, a reference from a different song of the same genre (27 pairs), the median tone distance to the reference goes from 11.1 to 3.6 dB with ours, 4.9 dB with Matchering and 9.5 dB with ITO-Master.

Loudness without damage. The loudest 30 s of each

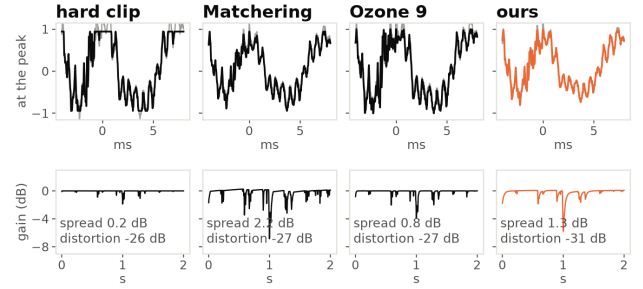


Figure 4: The same two seconds through four limiters, each driven to the loudness and true peak of the Ozone version. Top: the waveform at the largest peak (grey: the input scaled by the fitted gain). Bottom: the applied gain; “distortion” is the energy a smooth gain cannot explain, relative to the output.

MUSDB18-HQ test mixture (-14.7 LUFS, PLR 14.3 dB) is driven into each limiter until the output measures -9 LUFS, with the limiter’s own ceiling lowered until the $4\times$ oversampled peak respects -1 dBTP. We split what a limiter did into a smooth gain, fitted in 10 ms windows, and the remainder: *distortion* is the remainder relative to the signal, and *gain movement* is the 5th to 95th percentile spread of the fitted gain. A clipper scores badly on the first and a pumping limiter on the second, and Figure 4 shows both on one excerpt. On 50 mixes a hard clip measures -23.9 dB distortion at 0.8 dB of gain movement, Matchering 2.0 [23] -27.7 at 4.0, ffmpeg’s alimiter -37.1 at 6.0, our single stage -44.3 at 6.3, and the two-stage limiter -37.7 at 3.6 with a 1 dB clip stage or -32.7 at 2.3 with 2 dB, so the clip stage trades distortion for stillness. Left at their defaults all baselines overshoot the true-peak ceiling by 0.9 to 1.9 dB. Against the Ozone 9 Maximizer versions of the same songs [24], at the loudness and true peak of those versions (-7.5 LUFS), ours leaves 5 to 8 dB less distortion at equal or lower gain movement, while Ozone changes the long-term spectrum least (0.95 against 1.15 to 1.32 dB). These are signal measures of one preset of a product tuned by ear; which limiter sounds better is a question for a listening test.

4.5. Negative results

Predicting codec tokens. We first built the obvious alternative: encode with EnCodec [45] at 24 kbps, predict clean tokens with a 1.08 B parameter 1-D U-Net, decode. It cannot improve audio, and three measurements say why. A 1 dB gain change leaves 100% of tokens unchanged, because the codec normalizes each segment. Decoding the tokens of clean music gives 9.6 dB SI-SDR against the audio that was encoded, a ceiling below many damaged inputs. And losses on decoded audio pass through an arg max and carry no gradient. Probes detect degradations with AUROC 0.68 from tokens and 0.78 from continuous latents.

Blind equalization and per-stem tone. The controller’s equalization output stayed near zero for most of training and ends at a 14% reduction of tonal error, which is what the bound of Section 4.4 predicts and what Peladeau et al. [15] and Moliner et al. [16] would lead one to expect. Stems do not rescue this: correcting each separated stem toward the tone of its instrument class made every test case worse, with true stems as much as with separated ones, because instrument tone varies more between songs (5.5 dB per band for vocals) than the mix does.

De-compression and de-limiting. Specialists trained to undo compression and brickwall limiting with a full-rate gain head did not learn (envelope error 1.27 to 1.19 dB and 1.78 to 1.66 dB, while disturbing clean input by more). The released De-limiter [17] on clips limited with our limiter raises the crest factor from 10.1 to 12.6 dB (clean 16.6) and SI-SDR from 11.7 to 12.1 dB: some peaks come back, and it transfers only partly to a limiter it was not trained on. The pipeline measures dynamics and does not claim to undo them.

5. LIMITATIONS

All damage in our evaluation is synthetic, applied to real music; we have not evaluated recordings damaged in the field. On measured rooms outside the training set the dereverberation gain is 2.6 dB, half of the gain on training families. The tempo-grid rule leaves a quarter of added echoes in place. Soft saturation and noise are not improved, and clipping is detected only while its flat ceiling survives. The limiter comparison uses signal measures, and we have not run a listening test. The fine-tuned weights derive from a GPL-licensed checkpoint and part of the training data excludes commercial use.

6. CONCLUSION

A system for finished mixes does better when it measures before it acts. Exact DSP rebuilds clipped peaks and removes echo better than a network does, a masking transformer with a warm start removes reverberation without a codec ceiling, and a model trained to remove a fault turns out to be a meter for it. Measurements of released music draw the line between what mastering can decide blind (dynamics, loudness, peaks) and what needs a reference (tone and image). The audit on clean music set the thresholds; we recommend it.

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